

The water cycle

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Introduction

The water cycle is the continuous movement of water between the oceans, the air and the land. The total amount of water on Earth stays almost the same over time; what changes is where it is and in what form. Water moves as a liquid in rivers and rain, as a gas in the air, and as a solid in snow, glaciers and ice sheets, and it changes from one form to another as it gains or loses heat.

The Sun drives the cycle. Its energy warms the oceans and the land and turns liquid water into vapour, and gravity brings the water back down again as rain and snow and pulls it downhill towards the sea. About 97 per cent of all water is salt water in the oceans. Most of the fresh water is locked in ice caps and glaciers, and only a small fraction is in lakes, rivers and the air at any moment.

Evaporation

What evaporation is

Evaporation is the change of liquid water into water vapour, a gas. It happens at the surface of oceans, lakes, rivers and wet ground whenever some water molecules gain enough energy to escape into the air. Warm water evaporates faster than cold water, and wind speeds evaporation up because it carries away the moist air just above the surface and replaces it with drier air.

The oceans supply most of the water vapour in the air, roughly nine tenths of it. When sea water evaporates, the salt stays behind, so the vapour that rises is fresh water. This is why rain is fresh even though most of it began in the sea. Evaporation also cools the surface it leaves, because the escaping molecules carry heat away with them, the same effect that cools skin as sweat dries.

Transpiration and sublimation

Plants add water vapour to the air as well. They draw water up from the soil through their roots and release it through tiny pores in their leaves, a process called transpiration. A single large tree can release hundreds of litres of water on a hot day, and over a forest transpiration can supply as much vapour as a lake of the same size.

Ice and snow can also turn straight into vapour without melting first. This is called sublimation, and it happens most in dry, windy and sunny places, such as high mountains. Together, evaporation, transpiration and sublimation are the ways water enters the air.

Condensation

As warm, moist air rises it expands and cools. Cool air can hold less water vapour than warm air, and when the air cools to its dew point the vapour begins to condense back into tiny liquid droplets. The temperature at which this happens depends on how much vapour the air holds: the more humid the air, the higher its dew point.

Condensation needs a surface to form on. In the air, that surface is provided by tiny particles of dust, smoke, pollen and sea salt, called condensation nuclei. Billions of droplets formed around these particles make up a cloud. Fog is simply a cloud at ground level, and dew is the same process on cold grass or a window on a clear night. Condensation releases the heat that evaporation took in, and that heat warms the surrounding air and helps storms grow.

Precipitation

Rain and snow

Cloud droplets are far too small to fall. Precipitation begins when they collide and merge, or when ice crystals in a cold cloud grow at the expense of the droplets around them, until the drops or crystals are heavy enough to fall against the rising air. A raindrop holds the water of about a million cloud droplets.

Whether precipitation reaches the ground as rain or snow depends on the temperature of the air it falls through. Snow forms when ice crystals in a cloud below freezing grow and stick together into flakes. If the air near the ground stays below freezing the flakes arrive as snow; if they pass through a warm layer they melt and arrive as rain. Sleet forms when rain freezes again in a cold layer just above the ground.

Hail

Hail forms only in thunderstorms with strong updrafts. A small ice pellet is carried up and down through the cloud, and each trip through a layer of supercooled water adds a coat of ice. A hailstone that grows too heavy for the updraft to hold falls to the ground as ice. Cutting a large hailstone in half shows its rings, one for each layer it collected.

Collection

Runoff

Water that reaches the ground collects in several ways. Some runs off the surface into streams and rivers, which carry it back to lakes and the sea. Runoff is greatest where the ground is steep, already soaked or sealed by rock, concrete or frozen soil, and it is what causes rivers to rise after heavy rain.

Snow and ice store water for longer. A snowpack can hold the winter's precipitation until spring, when it melts and feeds rivers just as farms need the water most. Glaciers and ice sheets hold water for thousands of years, and most of the world's fresh water is stored in them.

Groundwater

Water that soaks into the soil instead of running off is called infiltration. Some of it is taken up by plant roots; the rest seeps down until it reaches a layer where the spaces in the rock are full of water. The top of that layer is the water table, and the water below it is groundwater.

Rock that holds groundwater and lets it flow is called an aquifer. Groundwater moves slowly, often only a few metres a year, and can stay underground for centuries before it returns to the surface in springs, rivers and wells. Wells that pump water out faster than rain can refill the aquifer cause the water table to fall.

People and the cycle

People change the water cycle in several ways. Cutting down forests reduces transpiration and can make a region drier, while cities sealed with concrete send more rain straight into drains as runoff instead of letting it soak in. Dams store river water in reservoirs, where a larger surface lets more of it evaporate.

A warmer climate speeds the cycle up. Warmer air can hold more water vapour, about seven per cent more for each degree Celsius, so the same storm can drop more rain. Warmer temperatures also melt glaciers and snowpacks earlier, which changes when rivers run high and low.